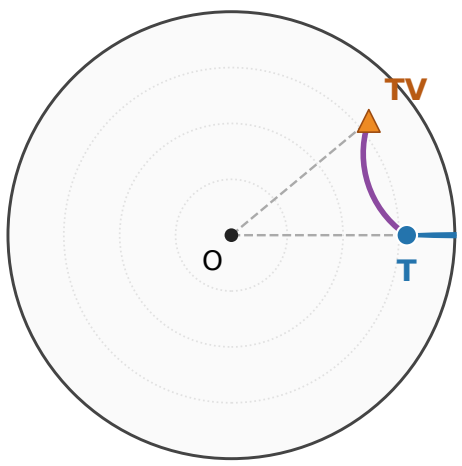
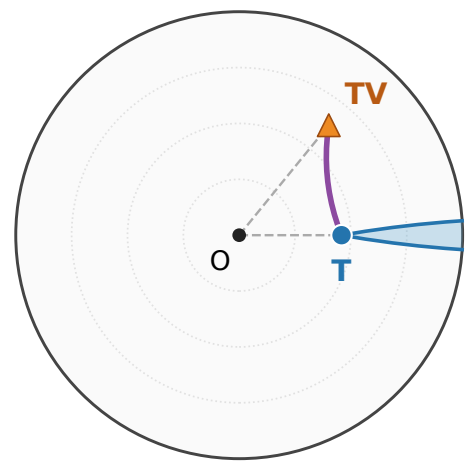
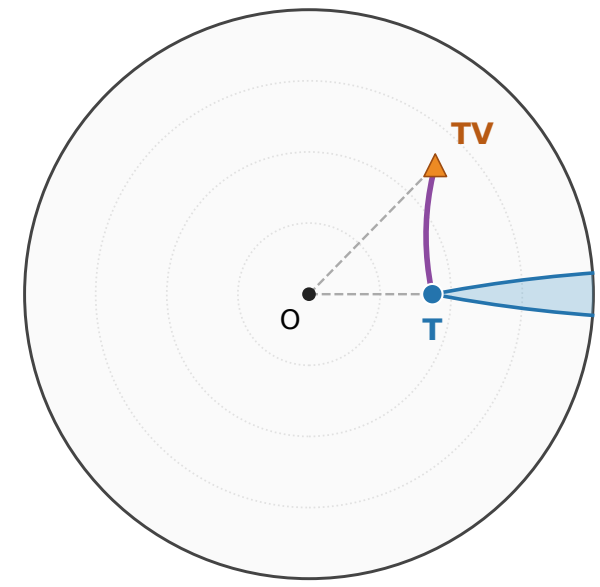
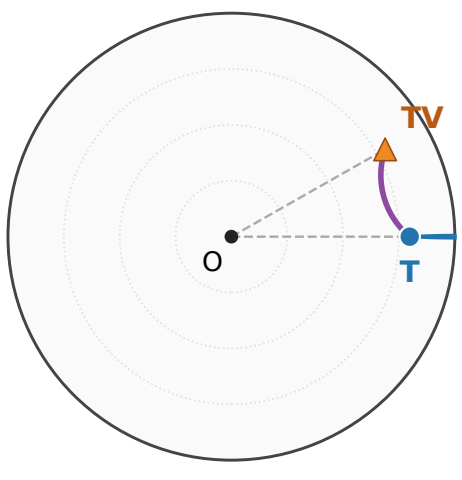
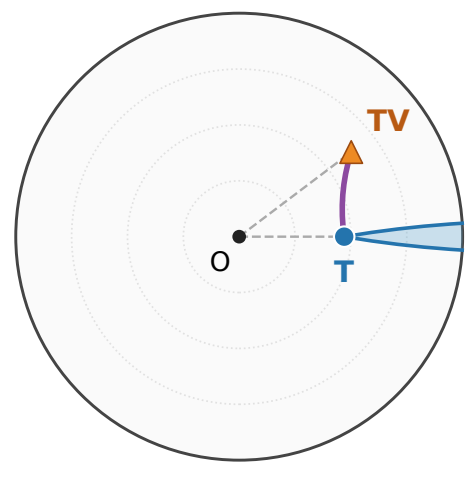
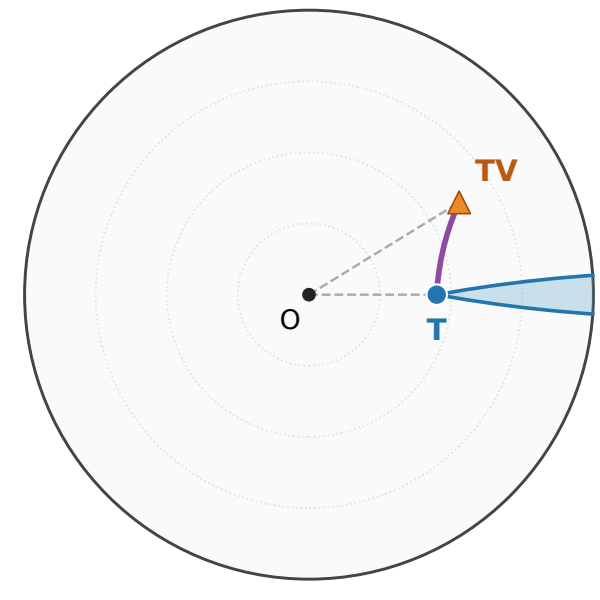
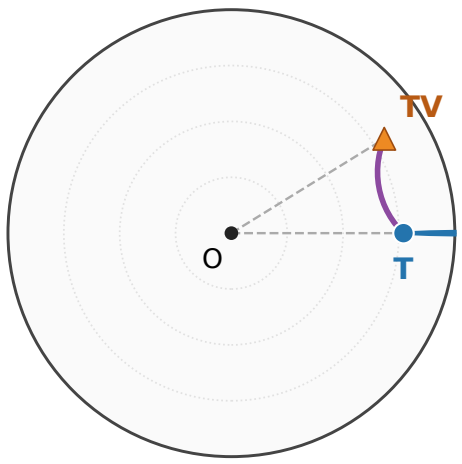
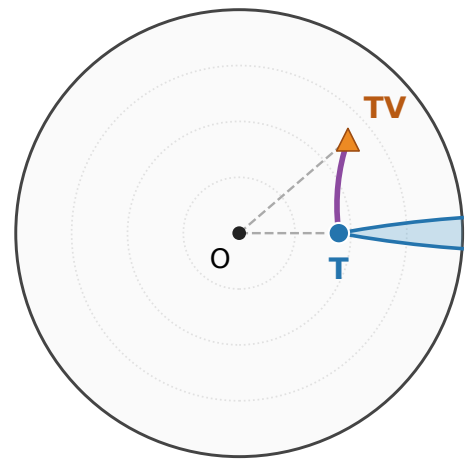
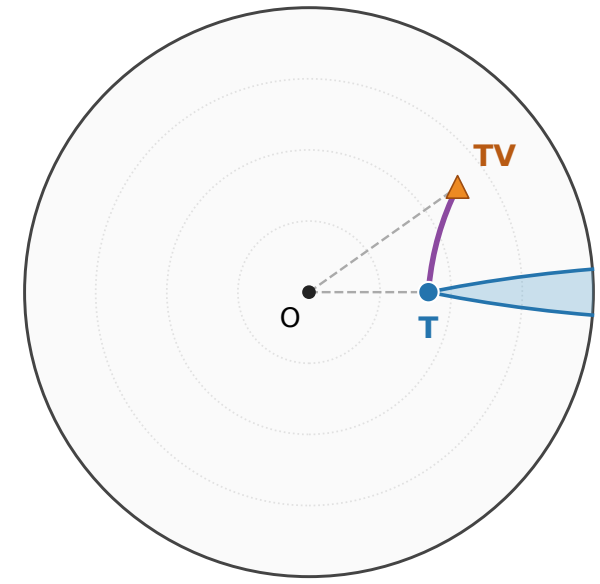
Actual Poincare-ball examples: T -> TV

Same three validation videos across models | real 512D features | epoch 30 | seed 50

Each panel is the exact O-parent-child hyperbolic plane, rotated so the parent is on the positive horizontal axis.

	Original C0 lambda=.1, c=1, tau=.07 Ball radius R=1.000	Current fixed lambda=.2, c=1, tau=.035 Ball radius R=1.000	Current learned lambda=.2, c=.617123, tau=.043651 Ball radius R=1.273
Sample 1 validation #1463	 r(T)=2.117 r(TV)=2.197 d(parent, child)=2.334 psi=2.80 deg phi=146.22 deg OUTSIDE cone; excess=143.42 deg	 r(T)=0.990 r(TV)=1.498 d(parent, child)=1.340 psi=9.93 deg phi=112.06 deg OUTSIDE cone; excess=102.13 deg	 r(T)=1.183 r(TV)=1.906 d(parent, child)=1.558 psi=10.79 deg phi=102.41 deg OUTSIDE cone; excess=91.62 deg
Sample 2 validation #1598	 r(T)=2.187 r(TV)=2.149 d(parent, child)=1.902 psi=2.61 deg phi=140.40 deg OUTSIDE cone; excess=137.80 deg	 r(T)=1.020 r(TV)=1.477 d(parent, child)=1.057 psi=9.54 deg phi=98.38 deg OUTSIDE cone; excess=88.84 deg	 r(T)=1.231 r(TV)=1.844 d(parent, child)=1.170 psi=10.24 deg phi=87.14 deg OUTSIDE cone; excess=76.90 deg
Sample 3 validation #1611	 r(T)=2.042 r(TV)=2.221 d(parent, child)=1.951 psi=3.03 deg phi=136.04 deg OUTSIDE cone; excess=133.01 deg	 r(T)=0.959 r(TV)=1.522 d(parent, child)=1.159 psi=10.35 deg phi=97.84 deg OUTSIDE cone; excess=87.49 deg	 r(T)=1.140 r(TV)=1.931 d(parent, child)=1.340 psi=11.31 deg phi=85.79 deg OUTSIDE cone; excess=74.48 deg

Parent embedding Child embedding Parent-child geodesic Parent entailment cone

Coordinates are curvature-aware Poincare coordinates; the joint-learning model has a larger ball because its c is smaller.

The parent-child geometry is preserved in each slice. Absolute orientations across different panels do not describe a global 2D embedding.